

OLED Evaporation — Process & Device Data Correlation Report

From co-evaporation process to device characteristics — correlations, FDC/SPC, part life

1. Overview

This report documents an interactive dashboard that explores the correlations among 15 signals obtainable on an OLED organic co-evaporation line — vacuum/residual-gas, source temperatures, QCM deposition rates, doping concentration, film thickness, and device metrics (driving voltage, efficiency, EQE, CIE color, lifetime). The data is synthetic but encodes real OLED physics (Arrhenius evaporation, doping->color/efficiency, thickness->voltage, residual water->lifetime); real EES/MES/FDC logs drop in directly.

2. Collected data signals (15)

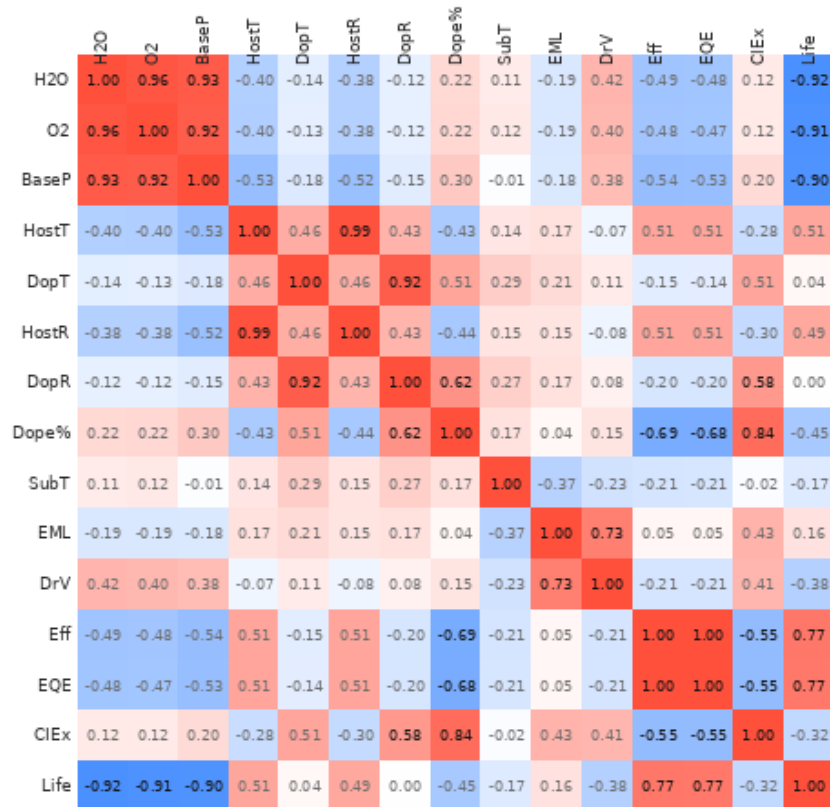
Group	Signal	Unit	Meaning
Vacuum/RGA	H2O partial	$\times 10^{-8}$ Torr	residual water - primary lifetime driver
	O2 partial	$\times 10^{-8}$ Torr	residual oxygen - degradation/dark spots
	Base Pressure	$\times 10^{-7}$ Torr	chamber vacuum / leak
Source	Host Temp	C	host cell temperature (Arrhenius)
	Dopant Temp	C	dopant cell temperature
Rate (QCM)	Host Rate	A/s	host deposition rate
	Dopant Rate	A/s	dopant deposition rate
Process	Doping %	%	doping conc. = $\text{Dop}/(\text{Host}+\text{Dop})$
	Substrate Temp	C	substrate temperature (morphology)
	EML thickness	nm	emission-layer thickness (shutter timing)
Device (QC)	Driving V	V	driving voltage
	Efficiency	cd/A	current efficiency
	EQE	%	external quantum efficiency
	CIE x	-	color coordinate x
	Lifetime LT95	h	time to 95% luminance

3. Physics models used to generate the data

- Arrhenius evaporation: $\text{rate} \sim \exp((T-T_0)/\text{scale})$; source temperature raises rate exponentially.
- Doping % = $\text{Dopant Rate} / (\text{Host Rate} + \text{Dopant Rate}) \times 100$. Host is PID-controlled, so doping tracks the dopant rate.
- Efficiency (cd/A): inverted-U peaking at the optimal doping (~6%); over-doping causes concentration quenching.
- EML thickness -> driving voltage: thicker organics raise the voltage; thickness also shifts CIE via microcavity.
- Contamination (residual H2O/O2) -> lifetime: more residual water/oxygen exponentially shortens LT95 (dark spots).

4. Correlation heatmap (Pearson r)

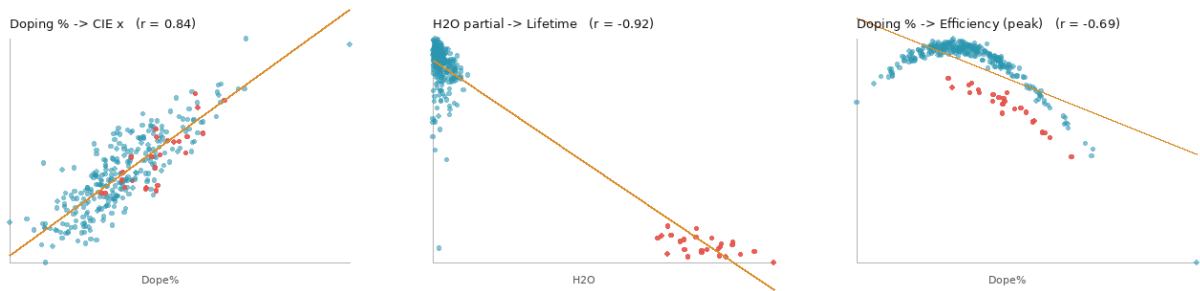
Color encodes r (blue -1, white 0, red +1). The vacuum/contamination cluster (H₂O, O₂, Base P) is mutually strongly positive; source-temperature->rate (Arrhenius) is ~1; doping->color is strongly positive; doping->efficiency is negative (over-doping side); and contamination->lifetime is strongly negative.



5. Key correlations and interpretation

Signal pair	r	Interpretation
Host Temp ~ Host Rate	+0.99	Arrhenius - temp up -> rate up (exp)
Dopant Temp ~ Dopant Rate	+0.92	dopant temp up -> dopant rate up
Dopant Rate ~ Doping %	+0.62	dopant rate up -> doping up
Doping % ~ CIE x	+0.84	doping up -> color shift (direct)
Doping % ~ Efficiency	-0.69	over-doping -> efficiency down (quenching)
EML thick ~ Driving V	+0.73	thickness up -> driving voltage up
EML thick ~ CIE x	+0.43	thickness -> microcavity color shift
H2O ~ Lifetime	-0.92	residual water up -> lifetime down (strong)
O2 ~ Lifetime	-0.91	residual oxygen up -> lifetime down
Efficiency ~ Lifetime	+0.77	efficiency up -> lifetime up (together)
Efficiency ~ EQE	+1.00	efficiency and EQE are essentially identical
H2O ~ O2	+0.96	contamination signals rise together

6. Representative scatter plots



- Higher doping shifts CIE x — color is directly set by doping (strong positive correlation).
- Higher residual water sharply reduces lifetime — contamination is the primary lifetime driver (strong negative).
- Efficiency peaks at the optimal doping (inverted-U) — invisible to linear r, revealed by the scatter.

7. Dashboard components

- Correlation heatmap: 15x15 Pearson r; click a cell -> scatter of that pair.
- Scatter: point cloud + regression line + r/r^2 ; contamination-event points in red.
- Time series: normalized key signals with the contamination event shaded.
- SPC chart: 3-sigma UCL/LCL on quality-critical values (doping/CIE); late doping drift breaches limits.
- FDC: residual-gas contamination anomaly score with ALARM.
- Part/source life: source running time, QCM crystal, material consumed, FMM cycles gauges.
- Causal diagram: source temp -> rate -> doping/thickness -> device metrics; contamination -> lifetime.

8. FDC, SPC and part life

FDC finds anomalies early from many sensors (here a residual H₂O/O₂ score). SPC standardizes recipe quality values (doping/CIE/thickness) with 3-sigma limits to ensure uniformity. Part life sizes the replacement cycle from cumulative source/crystal/mask usage. In this data a late doping drift (SPC breach) and a contamination event (FDC alarm) coincide, degrading efficiency, color and lifetime.

9. Correlation vs causation (caution)

Not every strong correlation is causal. If Substrate Temp and EML thickness happen to vary on similar periods, a spurious correlation can appear (de-phased here to weaken it). Read physical causation from the diagram, and use correlations as a starting point for hypotheses and anomaly detection.

10. Usage, limits and conclusion

On a real line, replace the synthetic data blocks with EES/MES/FDC logs of the same shape. The numbers here are model-based synthetic values, meant to show the structure of the relationships rather than absolute values. In short, the OLED process is a causal chain 'source temp -> rate -> doping/thickness -> device metrics; contamination -> lifetime', and the correlation heatmap is a powerful entry point to check that chain from data and tie it to FDC/SPC.